Euclid's Elements — Book I
Proposition 36

Formal Reconstruction — Technical Documentation

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Parallelograms which are on equal bases and in the same parallels equal one another.

1 Introduction

Proposition I.36 is the first immediate reuse of the content machinery introduced at I.35. Euclid passes from parallelograms on the same base to parallelograms on equal bases. The classical proof constructs an intermediate parallelogram and then applies I.35 twice.

The present reconstruction keeps the same content architecture but changes the geometric route to the intermediate parallelogram. The final Lean theorem does not import Proposition I.33 and does not require an explicit crossing point. Instead it reconstructs the missing pair of parallel sides from the developed Hilbert theory.

The conclusion remains synthetic. No numerical area is introduced. The two parallelogram terms are proved formally equicomplementable in the scissors calculus already used for I.35.

2 The Classical Configuration

Let $ABCD$ and $EFGH$ be parallelograms on equal bases BC and FG, lying between the same two parallels. In the standard ordered configuration the points satisfy

$$A - D - E - H, \qquad B - C - F - G.$$

The bases satisfy

$$BC \cong FG.$$

The current Lean theorem makes this order explicit through four strict betweenness hypotheses:

$$A - D - E, \qquad D - E - H, \qquad B - C - F, \qquad C - F - G.$$

Thus the formal statement is configuration-faithful to the standard ordered diagram, but it has stronger positional hypotheses than Euclid's order-free wording.

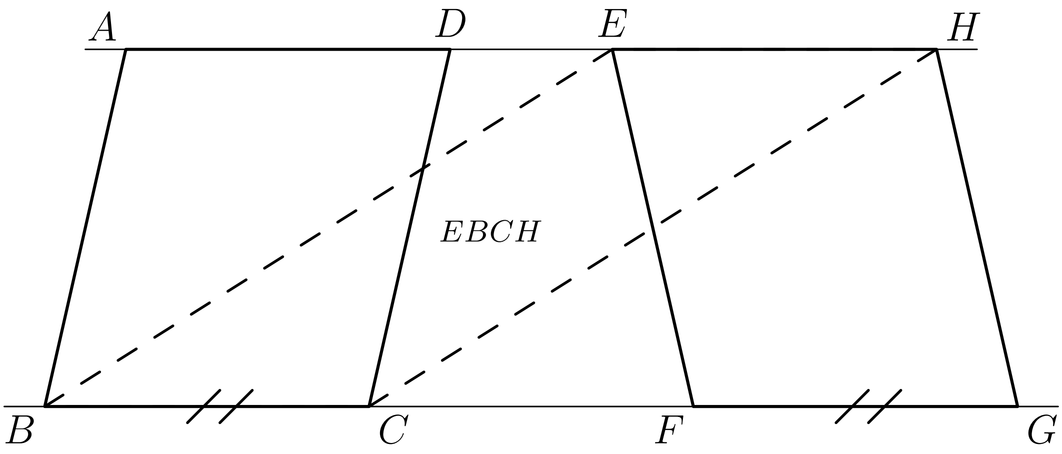

Figure 1: The ordered configuration used by the Lean theorem. The dashed quadrilateral $EBCH$ is the intermediate parallelogram constructed internally by the reconstruction.

3 Classical Proof

Euclid first uses Proposition I.34 on the parallelogram $EFGH$:

$$EH \cong FG.$$

Since the bases are equal,

$$BC \cong FG,$$

Common Notion 1 gives

$$BC \cong EH.$$

The segments BC and EH are also parallel. Proposition I.33 is then used to conclude that the joining segments EB and HC are equal and parallel, hence $EBCH$ is a parallelogram.

Now $ABCD$ and $EBCH$ are parallelograms on the same base BC and in the same parallels, so Proposition I.35 gives

$$ABCD = EBCH.$$

Similarly, $EFGH$ and $EBCH$ are equal by I.35. Common Notion 1 therefore yields

$$ABCD = EFGH.$$

The classical dependency graph is

$$\begin{gathered}
\text{I.34} + BC \cong FG \\
\Downarrow \\
BC \cong EH \\
\Downarrow \quad \text{I.33} \\
EBCH \text{ is a parallelogram} \\
\Downarrow \quad \text{I.35 twice} \\
ABCD = EBCH = EFGH \\
\Downarrow \quad \text{C.N.1} \\
ABCD = EFGH.
\end{gathered}$$

This is exactly the role of I.36 in the classical sequence: it is a generalization of I.35 obtained by inserting one intermediate parallelogram.

4 The Main Issue: Constructing the Intermediate Parallelogram

The difficult part of the formal reconstruction is not the content transitivity. Once $EBCH$ has been obtained, I.35 already supplies the required equality of figures. The real issue is geometric:

construct $EBCH$ from the ordered I.36 data.

The final proof does this without Proposition I.33 and without introducing a crossing witness.

4.1 The upper and lower parallel sides

The helper

```
i36_equal_parallel_EH_BC
```

extracts two facts:

$$EH \parallel BC, \qquad EH \cong BC.$$

Parallelism is transported from the side FG of $EFGH$ to the collinear carrier BC. Congruence comes from opposite-side congruence in the parallelogram together with the hypothesis $BC \cong FG$.

4.2 The shifted bottom segments

From

$$B - C - F - G, \qquad BC \cong FG,$$

the helper

```
i36_bottom_shift
```

derives

$$BF \cong CG$$

by segment additivity. This is the metric relation needed for the later SAS comparison.

4.3 The corresponding angle

The helper

```
i36_corresponding_angle
```

proves

$$\angle EFB \cong \angle HGC.$$

This is where the formal proof exposes the side-of-line data hidden by the ordinary diagram. A point X is constructed with

$$E - F - X.$$

The line through F and G is used as a carrier. The proof establishes the required SameSide and OppositeSide relations, transports $EF \parallel GH$ to $FX \parallel GH$, and then applies the developed Hilbert theorem

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

to obtain an alternate-angle congruence. Same-ray normalization and vertical angles convert that statement to the required corresponding-angle form.

4.4 SAS across the two bases

The helper

```
i36_cross_triangles_congruent
```

compares the triangles

$$\triangle FEB \qquad \text{and} \qquad \triangle GHC.$$

The three inputs are

$$FE \cong GH, \qquad FB \cong GC, \qquad \angle EFB \cong \angle HGC.$$

SAS therefore produces a full `TriangleCongruenceResult`. In particular,

$$\angle FBE \cong \angle GCH.$$

4.5 Recovering the second pair of parallel sides

The helper

```
i36_cross_parallel
```

uses the angle obtained from SAS to prove

$$EB \parallel HC.$$

Choose T with

$$B - T - C$$

and extend HC beyond C to a point Y:

$$H - C - Y.$$

The bottom carrier gives the required line-incidence data. Because E and H lie on the same side of that carrier, while H and Y lie on opposite sides, one obtains that E and Y are on opposite sides.

The angle from SAS is normalized to the rays through T and Y. Vertical angles at C give

$$\angle TBE \cong \angle TCY.$$

The neutral converse alternate-angle criterion

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

then yields

$$BE \parallel CY.$$

Finally parallelism is transported along the collinear triple H, C, Y:

$$EB \parallel HC.$$

The helper

```
i36_intermediate_EBCH_no_crossing
```

combines

$$EH \parallel BC \qquad \text{and} \qquad EB \parallel HC$$

and returns

$$\text{IsParallelogram}(E, B, C, H).$$

5 Proof Architecture of the Reconstruction

After separating geometry from content, the final proof has a short top-level shape.

The geometric branch is

$$\begin{gathered}
BC \cong FG, \quad EFGH, \quad B - C - F - G \\
\Downarrow \\
EH \parallel BC, \quad EH \cong BC \\
\Downarrow \\
BF \cong CG \\
\Downarrow \\
\angle EFB \cong \angle HGC \\
\Downarrow \quad \text{SAS} \\
\angle FBE \cong \angle GCH \\
\Downarrow \\
EB \parallel HC \\
\Downarrow \\
EBCH \text{ is a parallelogram.}
\end{gathered}$$

The content branch is

$$\begin{gathered}
ABCD,\ EBCH,\ EFGH \\
\Downarrow \quad \text{I.35} \\
ABCD \sim EBCH \\
\Downarrow \quad \text{I.35} \\
EBCH \sim EFGH \\
\Downarrow \quad \text{transitivity} \\
ABCD \sim EFGH.
\end{gathered}$$

Here $\sim$ denotes the formal `HilbertScissorsEquicomplementable` relation.

This separation is significant. The geometry of I.36 is new local work, but the equality-of-figures layer is inherited from I.35.

6 Lean Formalization

The current file imports only Proposition I.35:

```
import CGJteamLab.Proposition35
```

The public theorem is:

```
theorem euclid_proposition_36
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D E F G H : Geo.Point)
    (hABCD : IsParallelogram Geo A B C D)
    (hEFGH : IsParallelogram Geo E F G H)
    (hBCFG : Geo.Congruent B C F G)
    (hADE : Geo.Between A D E)
    (hDEH : Geo.Between D E H)
    (hBCF : Geo.Between B C F)
    (hCFG : Geo.Between C F G) :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertParallelogramTerm Geo E F G H)
```

The local helper sequence is:

```
i36_equal_parallel_EH_BC
i36_bottom_shift
i36_corresponding_angle
i36_cross_triangles_congruent
i36_cross_parallel
i36_intermediate_EBCH_no_crossing
i36_from_intermediate
euclid_proposition_36
```

The final theorem first derives the outer order information

$$A - E - H, \qquad A - D - H,$$

and

$$B - F - G, \qquad B - C - G$$

from the four explicit betweenness hypotheses. It then constructs $EBCH$ through the no-crossing helper and sends the result to the content helper.

The helper

```
i36_from_intermediate
```

is the exact content bridge. It applies I.35 once to $ABCD$ and $EBCH$. For the second application it reverses the parallelogram representations so that $EBCH$ and $EFGH$ are presented on the common base EH. The formal reversal is converted into scissors equivalence and then into equicomplementability. Two transitivity steps finish the proof.

7 Relationship with Earlier Propositions

7.1 Proposition I.33

I.33 is part of Euclid's classical DAG. After obtaining $BC \cong EH$ and $BC \parallel EH$, Euclid uses I.33 to conclude that the joining segments form the intermediate parallelogram $EBCH$.

The final Lean proof does not import Proposition I.33. Instead it proves $EB \parallel HC$ directly by the SAS and alternate-angle route described above. Thus I.33 remains historically relevant but is not a formal dependency of the final file.

7.2 Proposition I.34

Classically, I.34 supplies

$$EH \cong FG.$$

The current Lean proof accesses this content through the developed parallelogram API, in particular:

```
ParallelogramOppositeSidesCongruent
```

Thus the mathematical input of I.34 is present even though the Book I theorem is not called by name.

7.3 Proposition I.35

I.35 is the decisive content dependency. Once $EBCH$ has been constructed, the final result is obtained by two applications of I.35 plus transitivity of equicomplementability.

This agrees with the classical organization and with modern content-based treatments of I.36.

8 Hilbert and Book Zero Components Used

The proof uses three layers of developed infrastructure.

First, the Hilbert geometry layer supplies:

- parallelogram opposite-side congruence;
- transport of parallelism along collinear carriers;
- betweenness extension and inner/outer transitivity;
- SameRay normalization of angles;

- SameSide and OppositeSide transport;
- vertical angles;
- SAS triangle congruence;
- the Euclidean theorem giving equal alternate angles from parallel lines;
- the neutral converse criterion giving parallel lines from equal alternate angles.

Second, the Book Zero layer supplies elementary order and congruence infrastructure used by the developed Hilbert interface. Proposition I.36 does not introduce a new Book Zero theorem.

Third, the scissors layer inherited from I.35 supplies:

- parallelogram terms;
- representation reversal;
- conversion from scissors equivalence to equicomplementability;
- transitivity of equicomplementability.

No polygonal realization layer is introduced. The existing `HilbertScissorsTerm` representation is sufficient.

9 Formalization Notes

9.1 Geometry, content, and representation

I.36 is a useful example of the three-way separation introduced after I.35.

Geometry. The new work is the intrinsic construction of $EBCH$ from equal bases, parallel carriers, and the ordered configuration.

Content. No new equality-of-figures principle is required. I.35 and transitivity are sufficient.

Representation. The only extra bookkeeping is reversal of parallelogram terms before the second application of I.35. This is handled by existing scissors-equivalence helpers.

9.2 Hidden configuration data

The formal theorem exposes several facts left implicit by the classical diagram:

- the upper order $A - D - E - H$;
- the lower order $B - C - F - G$;
- the carrier line through F and G;
- SameSide information for E and H relative to that carrier;

- the extension point X with $E - F - X$;
- the interior point T with $B - T - C$;
- the extension point Y with $H - C - Y$;
- the required non-collinearity conditions for genuine angles;
- the OppositeSide condition used by the alternate-angle criterion.

These data are derived or constructed internally except for the four betweenness hypotheses in the theorem statement.

9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The local geometric route is informative about where the parallel axiom is used. The helper proving the corresponding angle invokes

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

which is the Euclidean direction

$$\text{parallel lines} \Longrightarrow \text{equal alternate angles.}$$

By contrast, the final step that recovers $EB \parallel HC$ invokes

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

which is the neutral converse direction.

Hilbert's Chapter IV theory of plane area works over Groups I–IV and does not require the continuity group. The formal reconstruction likewise introduces no continuity axiom at I.36.

9.4 Axiom audit

The final audit is:

```
'Geometry.euclid_proposition_36' depends on axioms:
[propext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

The geometric helpers have the smaller audit:

```
i36_cross_parallel:
[propext, Classical.choice, Quot.sound]

i36_intermediate_EBCH_no_crossing:
[propext, Classical.choice, Quot.sound]
```

The local project axiom

```
bookZero_nullSegment1
```

enters only through Proposition I.35 and hence through the content helper. The new I.36 geometry introduces no additional local axiom.

9.5 Source audit

The source audit gives a consistent picture.

Euclid/Joyce and Wilkins give the classical DAG

$$\text{I.34} \to \text{I.33} \to \text{I.35 twice} \to \text{C.N.1.}$$

Hilbert's Chapter IV, Section 19, Theorem 44 identifies equicomplementability of parallelograms with equal bases and equal altitudes as the correct synthetic area-theoretic target. It also warns against silently strengthening the result to unconditional equidecomposability in non-Archimedean settings.

Hartshorne's Section 22 states that I.36 follows from transitivity of equal content. The additional property commonly denoted (Z) is not required for I.36; it first becomes relevant in the discussion of I.37.

The Beeson proof corpus was used as an audit of hidden orientation and crossing data. The final Lean proof deliberately does not reproduce its case split: the intermediate parallelogram is derived intrinsically instead.

Borsuk–Szmielew and Greenberg remain control sources for order, parallelograms, neutral geometry, and the neutral/Euclidean boundary, but they do not determine the content architecture of the final proof.

10 Dependency Summary

The complete formal dependency chain is

$$\begin{gathered}\text{Hilbert axioms}\\ \Downarrow\\ \text{Hilbert / Book Zero order, congruence, side-of-line theory}\\ \Downarrow\\ \text{parallelogram and parallel-angle infrastructure}\\ \Downarrow\\ \text{I.36 geometric helpers}\\ \Downarrow\\ EBCH \text{ is a parallelogram}\\ \Downarrow\\ \text{I.35 twice + scissors representation normalization}\\ \Downarrow\\ \text{equicomplementability transitivity}\\ \Downarrow\\ \texttt{euclid\_proposition\_36}.\end{gathered}$$

The short comparison is

$$\begin{aligned}\text{Euclid} &: \text{I.34} \to \text{I.33} \to \text{I.35 twice},\\ \text{Lean} &: \text{equal-base geometry} \to \text{SAS} \to \text{alternate-angle parallelism} \to EBCH \to \text{I.35 twice}.\end{aligned}$$

New axiom:	no
New proposition-local helpers:	yes
New Book Zero theorem:	no
New polygon/content representation:	no
Build clean:	yes
`sorry`:	no

11 Conclusion

Proposition I.36 confirms that the scissors layer introduced at I.35 is already reusable. No new notion of area, content, or polygon is needed. The content argument is exactly the expected transitivity pattern through an intermediate parallelogram.

The principal formal contribution lies instead in the geometry of that intermediate figure. The final reconstruction removes both an explicit crossing witness and the Book I dependency on I.33. Equal-base information, parallel-line angle theory, SAS, side-of-line control, and a neutral alternate-angle criterion are sufficient to recover $EBCH$ directly.

Thus I.36 separates cleanly into a new geometric normalization stage followed by an already established I.35 content stage.